DUBLIN CITY UNIVERSITY

SEMESTER 1 EXAMINATIONS 2017/2018

MODULE: CA218 - Introduction to Databases

PROGRAMME(S):

BSc in Computer Applications (Sft.Eng.)

YEAR OF STUDY: 2

EXAMINERS: Mark Roantree (Ph:5636)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

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The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper (Please mark (X) as appropriate)

- [ ] *Log Tables*
- [ ] *Graph Paper*
- [ ] *Dictionaries*
- [ ] *Statistical Tables*
- [ ] *Thermodynamic Tables*
- [ ] *Actuarial Tables*
- [ ] *MCQ Only - Do not publish*
- [ ] *Attached Answer Sheet*

QUESTION 1 **[Total marks: 25]**

Entity Relationship Modelling

You must design a database for a Sports Centre in Dublin which has a gymnasium, swimming pool, basketball and squash courts. Thus, the **SportsCentre** operates four types of **Activity**. It also has 3 different membership types: pool only, gym only, or all 4 activities.

1(a) [10 Marks]

Create an Entity-Relationship diagram for this application where each entity has a minimum of 4 attributes. Each entity should have at least one relationship so that all entities are connected to at least one other entity. Use different types of relationships in your diagram. State clearly any assumptions made.

1(b) [7 Marks]

What type of relationship connects **Member** and **Activity**? Why is this type of relationship necessary in your example? When transforming from an E-R model to a relational schema, how would you manage this relationship?

1(c) [8 Marks]

What is meant by a *participation* constraint? Provide a description of both types of participation constraint. In your answer provide an example from the gymnasium case study of each type of participation constraint.

[End Question 1]

QUESTION 2 **[Total marks: 25]**

Database Environment

2(a) [6 Marks]

Explain the role of both the *conceptual* level and the *external* level in the 3-level schema architecture.

2(b) [10 Marks]

Explain why *schema mappings* are necessary in the 3-level schema architecture. Provide a detailed example (with sample schemas) of how these mappings operate with respect to the conceptual and external schemas.

2(c) [9 Marks]

We discussed 10 functions of a Database Management System in class. List 9 of these functions and provide a brief description of each.

[End Question 2]

QUESTION 3 ***[Total marks: 25]***

SQL Expressions
A recruitment company has the following schema:
Staff (staffNo, fname, lname, dept, skillCode)
Skill (skillCode, description, HourlyRate)
Project (projectNo, startDate, endDate, budget, projectManagerStaffNo)
Booking (staffNo, projectNo, dateWorkedOn, timeWorkedOn)

where:
Staff contains staff details and staffNo is the key.
Skill contains descriptions of skill codes (e.g. Programmer, Analyst, Manager, etc.) and the hourly rate for that skill; the key is skillCode.
Project contains project details and projectNo is the key.
Booking contains details of the date and the number of hours that a member of staff worked on a project and the key is (staffNo,projectNo).

3(a) [2 Marks]

List the names and departments of all employees whose skillCode is S12.

3(b) [4 Marks]

List all projects that have a maximum of 5 staff working on that project.

3(c) [5 Marks]

List all staff with an hourly rate greater than the average hourly rate.

3(d) [7 Marks]

Produce a complete list of all managers who are due to retire this year (at age 65), in alphabetical order of surname.

3(e) [7 Marks]

For all projects that were active in November 2017, list the staff name, project number and the date and number of hours worked on the project. Order results by staff name, within staff name by the project number, and within project number by date.

[End Question 3]

QUESTION 4 ***[Total marks: 25]***

Relational Algebra

4(a) [12 Marks]

Assume we have two relations R(a,b,c) and S(a,b,c), where R contains N_r tuples and S contains N_s tuples ($N_s > N_r > 0$). In other words, both R and S are identical in structure, neither relation is empty and S contains more tuples than R.

Provide the **minimum** *and* **maximum** cardinality for the result relation for each of the following relational algebra expressions and in each case, state any assumptions about the schemas that explain your answer.

(i) $R \cup S$
(ii) $R \cap S$
(iii) $R - S$
(iv) $R \times S$
(v) $\sigma_{a=1}\ ^{(R)}$
(vi) $\pi_a\ ^{(R)}$

4(b) [13 Marks]

Convert each of the following expressions to its equivalent in Relation Algebra.
(i) SELECT fname, lname FROM STUDENT where SID = '5125'
(ii) select the tuples for all students who either take CA119 and score over 80% or take CA218 and score over 70%. (Use a single expression)
(iii) Rewrite query (ii) using an intermediate result set.

[End Question 4]

QUESTION 5 *[Total marks: 25]*

Normalisation

staffNo	branchNo	BranchAddress	name	position	hoursPer Week
S4555	B002	5a Drumcondra Road East, Dublin 9.	Katie Taylor	Assistant	16
S4555	B002	5a Drumcondra Road East, Dublin 9.	Katie Taylor	Assistant	9
S4612	B004	10 Collins Ave. West, Dublin 9.	James McCarthy	Assistant	14
S4612	B004	10 Collins Ave. West, Dublin 9.	James McCarthy	Assistant	10

Figure 1: Hours Worked Table

The table shown in Figure 1 represents the hours worked per week for temporary staff at every branch for one company.

5(a) [6 Marks]

The table in Figure 1 is susceptible to update anomalies. Provide examples of how insertion, deletion, and modification anomalies could occur on this table.

5(b) [8 Marks]

Identify the functional dependencies present in the data shown in Figure 1. State any assumptions you make about the data.

5(c) [11 Marks]

Using the functional dependencies identified in part (b), describe the process of normalization by converting Table 1 to Third Normal Form (3NF) relations. Identify the primary and foreign keys in your 3NF relations.

[End Question 5]

[END OF EXAM]